

```

1 frames = int(input("Enter number of frames: "))
2 pages = list(map(int, input("Enter page reference string: ").split()))
3 memory = []
4 page_faults = 0
5 for i in range(len(pages)):
6     page = pages[i]
7     if page not in memory:
8         page_faults += 1
9
10        if len(memory) < frames:
11            memory.append(page)
12        else:
13            future = pages[i+1:]
14            replace_index = -1
15            farthest = -1
16
17            for j in range(len(memory)):
18                if memory[j] not in future:
19                    replace_index = j
20                    break
21            else:
22                pos = future.index(memory[j])
23                if pos > farthest:
24                    farthest = pos
25                    replace_index = j
26
27            memory[replace_index] = page
28
29        print("Page:", page, "->", memory)
30    print("Total Page Faults =", page_faults)

```

```

Enter number of frames: 6
Enter page reference string: 8
Page: 8 -> [8]
Total Page Faults = 1

Process finished with exit code 0

```